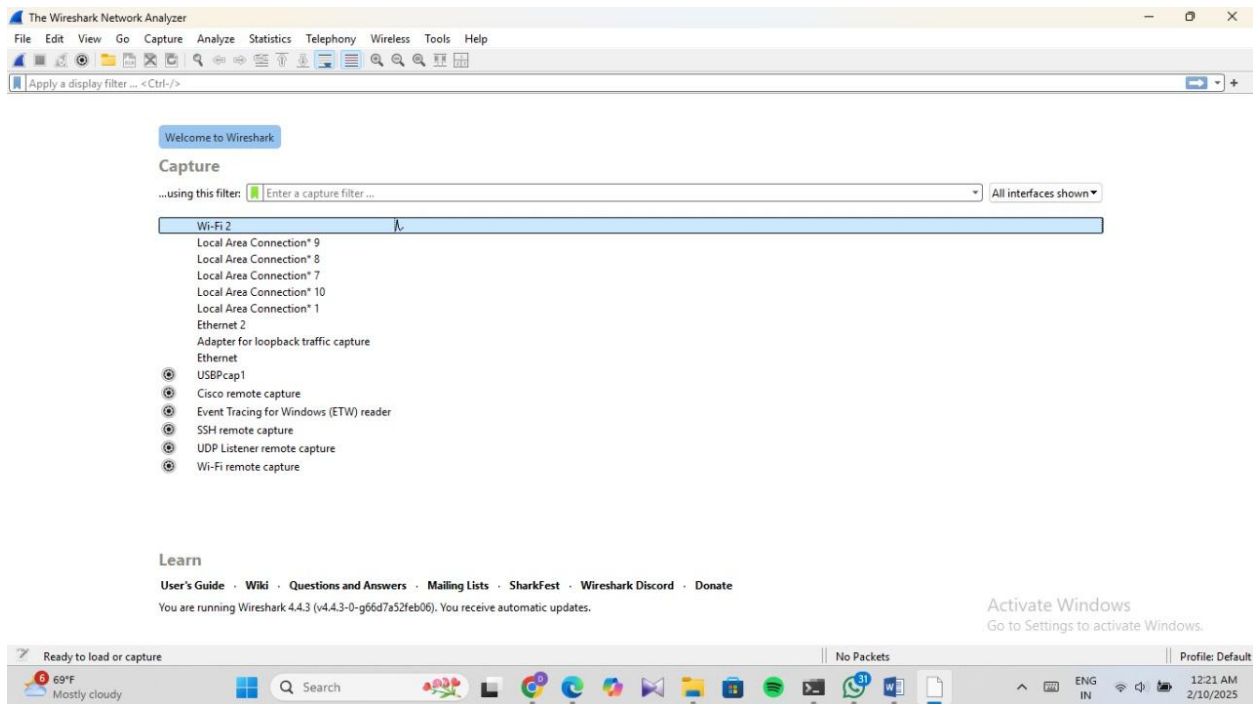


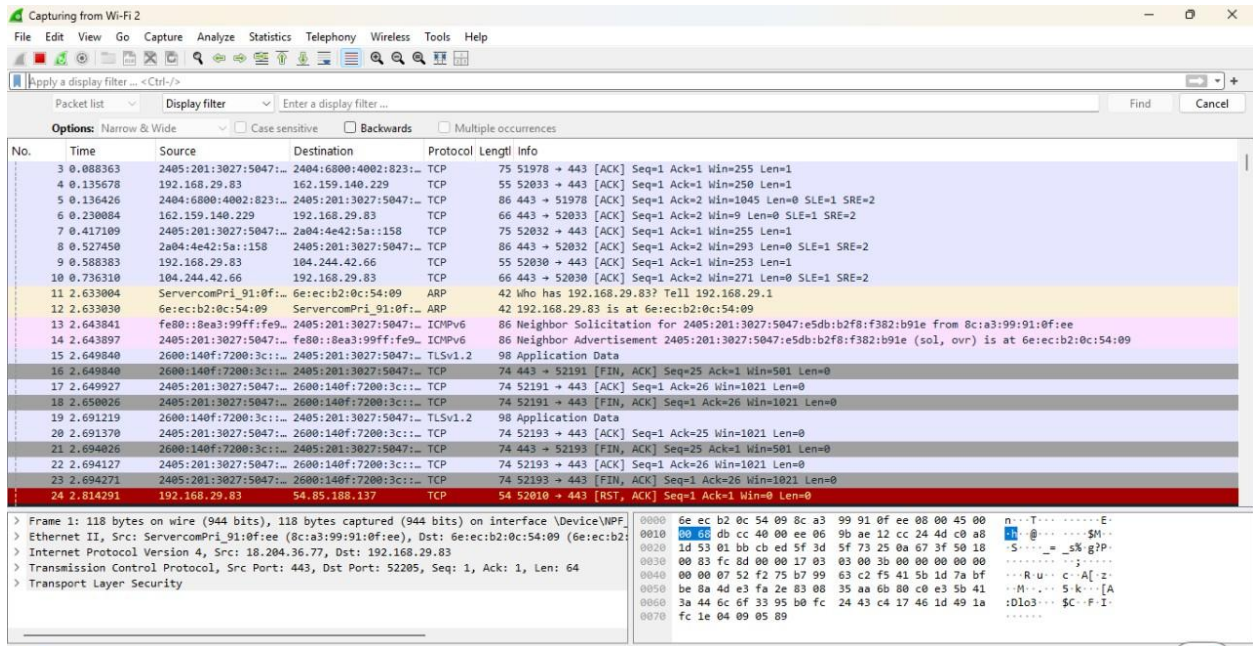
Wireshark

- ☒ Wireshark is a tool that captures and analyzes network traffic.
- ☒ It helps you see what data is being sent and received over a network.
- ☒ You can check network security, troubleshoot issues, and analyze packets.
- ☒ Wireshark is a network analysis tool that captures network traffic in real time and allows its analysis.
- ☒ It is used for troubleshooting network issues, checking security, and analyzing packets

Wireshark Installation



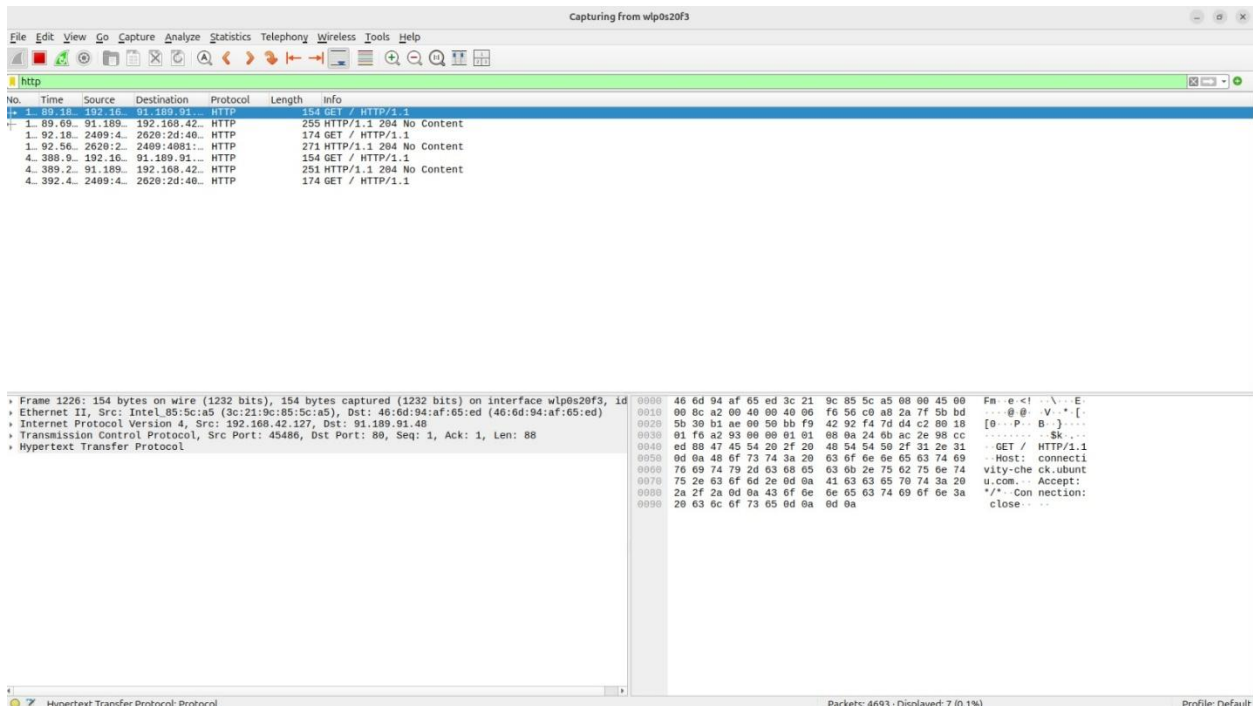
Capture packets while browsing the internet:



The image shows the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons for packet capture and analysis. The main window is divided into three panes: a packet list on the left, a packet details pane in the middle, and a packet bytes pane on the right. The packet list shows a list of captured packets with columns for No., Time, Source, Destination, Protocol, Length, and Info. The packet details pane shows the selected packet's structure, including Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. The packet bytes pane shows the raw data of the selected packet in hexadecimal and ASCII.

2. Identify HTTP Vs HTTPS traffic:

Identify HTTP traffic using http.host:



The image shows the Wireshark network protocol analyzer interface with a filter applied to capture HTTP traffic. The filter bar at the top shows 'http'. The packet list on the left shows a list of captured packets, with the first packet selected. The packet details pane in the middle shows the selected packet's structure, including Ethernet II, Internet Protocol Version 4, and Hypertext Transfer Protocol. The packet bytes pane on the right shows the raw data of the selected packet in hexadecimal and ASCII. The bottom status bar indicates that 4693 packets are displayed, representing 7.01% of the total capture.

Wi-Fi 2

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http.host

Options: Narrow & Wide Case sensitive Backwards Multiple occurrences

No.	Time	Source	Destination	Protocol	Length	Info
276	26.862063	192.168.29.31	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1
2272	165.887739	2405:201:3027:5047::...	2600:140f:c00:b854::...	HTTP	186	GET /connecttest.txt HTTP/1.1
2282	165.914097	192.168.29.83	184.85.220.72	HTTP	165	GET /connecttest.txt HTTP/1.1
3763	254.321825	192.168.29.47	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1
3774	256.764738	192.168.29.47	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1
3953	271.409519	192.168.29.47	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1
4002	281.581257	192.168.29.47	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1
4116	291.568648	192.168.29.47	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1
4302	301.519477	192.168.29.47	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1
4685	311.640174	192.168.29.47	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1
4715	321.522396	192.168.29.47	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1
4747	331.634852	192.168.29.47	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1
4946	341.526237	192.168.29.47	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1
5269	351.575662	192.168.29.47	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1
5342	361.529857	192.168.29.47	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1
5393	371.599285	192.168.29.47	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1
5507	381.512173	192.168.29.47	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1
5539	391.531886	192.168.29.47	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1
5946	455.535911	192.168.29.47	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1
6243	515.529928	192.168.29.47	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1
6638	575.524067	192.168.29.47	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1
7198	635.571488	192.168.29.47	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1

> Frame 3050: 167 bytes on wire (1336 bits), 167 bytes captured (1336 bits) on interface \Device\NPF...
> Ethernet II, Src: ce:fa:19:2f:85:d9 (ce:fa:19:2f:85:d9), Dst: 6e:ec:b2:0c:54:09 (6e:ec:b2:0c:54:09)
> Internet Protocol Version 4, Src: 192.168.29.31, Dst: 239.255.255.250
> User Datagram Protocol, Src Port: 54141, Dst Port: 1900
> Simple Service Discovery Protocol

Host Character string

Packets: 74352 - Displayed: 34 (0.0%) - Dropped: 0 (0.0%) Profile: Default

Identify HTTP Traffic using port 80:

Capturing from wlp0s20f3

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.port == 80

No.	Time	Source	Destination	Protocol	Length	Info
1.	92.18...	2020:2d:4002:1::190	2409:4081:8792:4b2a:b200:b099:49ba:616	TCP	94	80 -- 56754 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1370 SACK_PERM TSval=...
1.	92.56...	2020:2d:4002:1::190	2409:4081:8792:4b2a:b200:b099:49ba:616	HTTP	271	HTTP/1.1 204 No Content
1.	92.56...	2020:2d:4002:1::190	2409:4081:8792:4b2a:b200:b099:49ba:616	TCP	86	80 -- 56754 [FIN, ACK] Seq=186 Ack=89 Win=65536 Len=0 TSval=1160363959 TSecr=...
1.	92.97...	2020:2d:4002:1::190	2409:4081:8792:4b2a:b200:b099:49ba:616	TCP	86	80 -- 56754 [ACK] Seq=187 Ack=90 Win=65536 Len=0 TSval=1160363929 TSecr=763...
4.	392.4...	2020:2d:4000:1::98	2409:4081:8792:4b2a:b200:b099:49ba:616	TCP	94	80 -- 48962 [SYN, ACK] Seq=0 Ack=1 Win=64200 Len=0 MSS=1370 SACK_PERM TSval=...
4.	392.9...	2020:2d:4000:1::98	2409:4081:8792:4b2a:b200:b099:49ba:616	TCP	86	[TCP Previous segment not captured] 80 -- 48962 [FIN, ACK] Seq=186 Ack=89 W...
4.	392.9...	2020:2d:4000:1::98	2409:4081:8792:4b2a:b200:b099:49ba:616	TCP	271	[TCP Out-Of-Order] 80 -- 48962 [PSH, ACK] Seq=1 Ack=89 Win=65536 Len=185 TS...
4.	394.9...	2020:2d:4000:1::98	2409:4081:8792:4b2a:b200:b099:49ba:616	TCP	74	80 -- 48962 [RST] Seq=187 Win=0 Len=0
8.	692.1...	2020:2d:4000:1::2a	2409:4081:8792:4b2a:b200:b099:49ba:616	TCP	94	80 -- 58640 [SYN, ACK] Seq=0 Ack=1 Win=64200 Len=0 MSS=1370 SACK_PERM TSval=...
8.	692.8...	2020:2d:4000:1::2a	2409:4081:8792:4b2a:b200:b099:49ba:616	TCP	86	[TCP Previous segment not captured] 80 -- 58640 [FIN, ACK] Seq=190 Ack=89 W...
8.	692.8...	2020:2d:4000:1::2a	2409:4081:8792:4b2a:b200:b099:49ba:616	TCP	275	[TCP Out-Of-Order] 80 -- 58640 [PSH, ACK] Seq=1 Ack=89 Win=65536 Len=189 TS...
8.	693.4...	2020:2d:4000:1::2a	2409:4081:8792:4b2a:b200:b099:49ba:616	TCP	86	[TCP Retransmission] 80 -- 58640 [FIN, ACK] Seq=190 Ack=89 Win=65536 Len=0 ...
8.	694.5...	2020:2d:4000:1::2a	2409:4081:8792:4b2a:b200:b099:49ba:616	TCP	86	80 -- 58640 [ACK] Seq=191 Ack=90 Win=65536 Len=0 TSval=1160363959 TSecr=...
8.	857.3...	2404:6800:4009:82a::200e	2409:4081:8792:4b2a:b200:b099:49ba:616	TCP	94	80 -- 36820 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1370 SACK_PERM TSval=...
8.	857.3...	2404:6800:4009:82a::200e	2409:4081:8792:4b2a:b200:b099:49ba:616	TCP	86	80 -- 36820 [ACK] Seq=1 Ack=90 Win=269568 Len=0 TSval=31145379 TSecr=28169...
8.	857.4...	2404:6800:4009:82a::200e	2409:4081:8792:4b2a:b200:b099:49ba:616	TCP	1221	80 -- 36820 [PSH, ACK] Seq=1 Ack=402 Win=269568 Len=1135 TSval=31145442 TSe...
8.	857.4...	2404:6800:4009:82a::200e	2409:4081:8792:4b2a:b200:b099:49ba:616	HTTP/JSON	106	HTTP/1.1 200 OK, JSON (Application/json)
8.	857.5...	2404:6800:4009:82a::200e	2409:4081:8792:4b2a:b200:b099:49ba:616	TCP	86	80 -- 36820 [FIN, ACK] Seq=1156 Ack=403 Win=269568 Len=0 TSval=31145509 TSe...
9.	992.0...	2020:2d:4000:1::22	2409:4081:8792:4b2a:b200:b099:49ba:616	TCP	94	80 -- 58714 [SYN, ACK] Seq=0 Ack=1 Win=64200 Len=0 MSS=1370 SACK_PERM TSval=...
9.	992.6...	2020:2d:4000:1::22	2409:4081:8792:4b2a:b200:b099:49ba:616	TCP	86	[TCP Previous segment not captured] 80 -- 58714 [FIN, ACK] Seq=190 Ack=89 W...
9.	992.6...	2020:2d:4000:1::22	2409:4081:8792:4b2a:b200:b099:49ba:616	TCP	275	[TCP Out-Of-Order] 80 -- 58714 [PSH, ACK] Seq=1 Ack=89 Win=65536 Len=189 TS...
9.	993.2...	2020:2d:4000:1::22	2409:4081:8792:4b2a:b200:b099:49ba:616	TCP	86	80 -- 58714 [ACK] Seq=191 Ack=90 Win=65536 Len=0 TSval=3559619589 TSecr=389...
9.	991.6...	2409:4081:8792:4b2a:b200:b099:49ba:616	2620:2d:4000:1::22	TCP	94	58714 -- 80 [SYN] Seq=0 Win=64952 Len=0 MSS=1412 SACK_PERM TSval=3896508188 ...
9.	992.0...	2409:4081:8792:4b2a:b200:b099:49ba:616	2620:2d:4000:1::22	TCP	86	58714 -- 80 [ACK] Seq=1 Ack=1 Win=65024 Len=0 TSval=3896508066 TSecr=3559618...
9.	992.0...	2409:4081:8792:4b2a:b200:b099:49ba:616	2620:2d:4000:1::22	HTTP	174	GET / HTTP/1.1

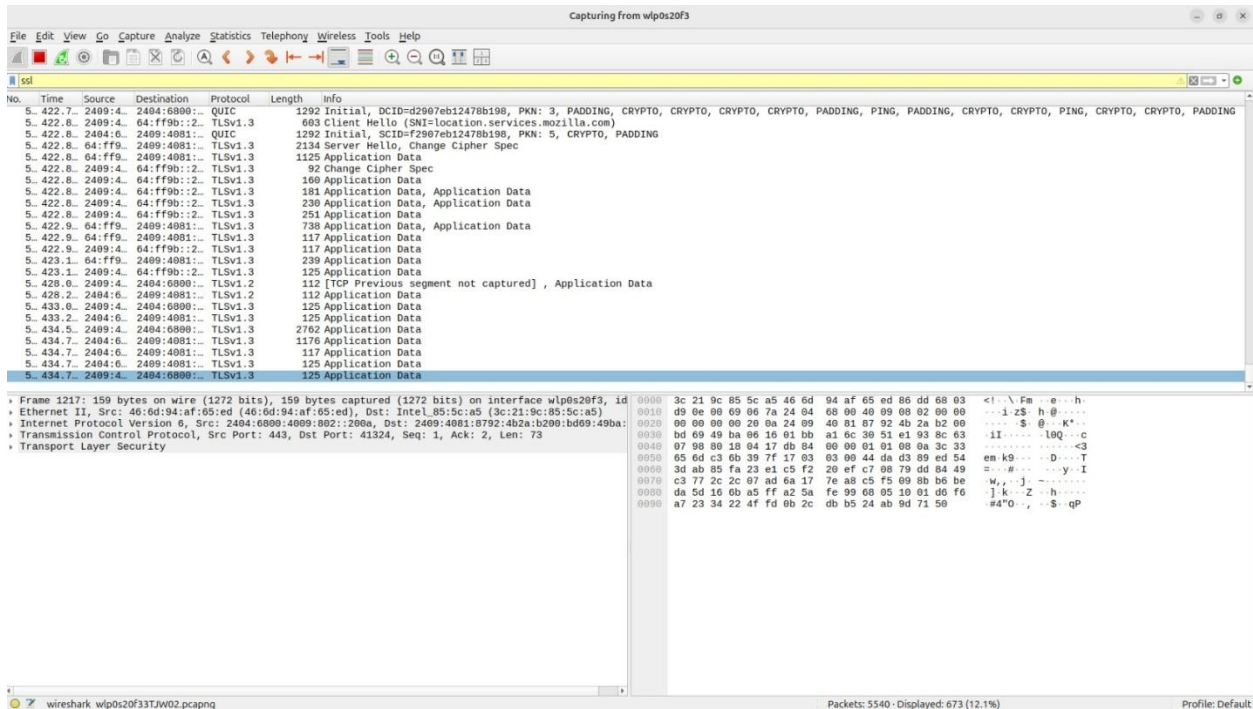
> Frame 8316: 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface wlp0s20f3, id 0
> Ethernet II, Src: 46:0d:94:af:05:ed (46:0d:94:af:05:ed), Dst: Intel_85:5c:a5 (3c:21:9c:85:5c:a5)
> Internet Protocol Version 6, Src: 2620:2d:4000:1::2a, Dst: 2409:4081:8792:4b2a:b200:b099:49ba:616
> Transmission Control Protocol, Src Port: 80, Dst Port: 58646, Seq: 191, Ack: 90, Len: 0

wireshark wlp0s20f33TJW02.pcapng

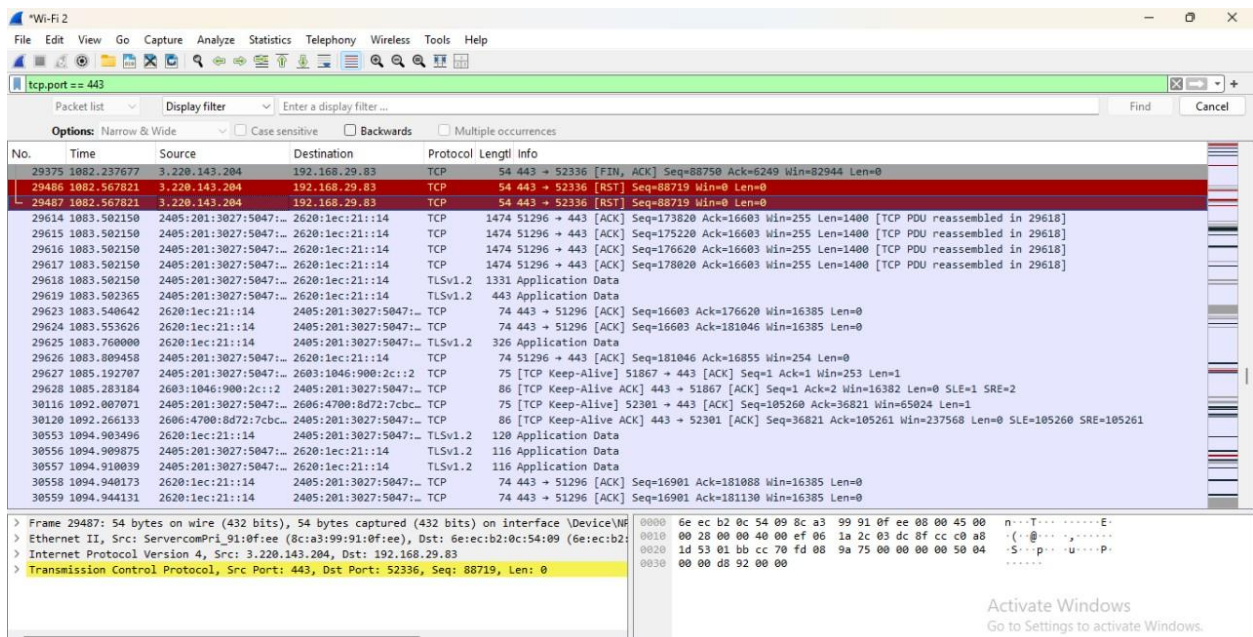
Packets: 10927 - Displayed: 92 (0.8%) Profile: Default

🔑 Identify HTTPS TRAFFIC:

💡 Using SSL filter:



💡 Using port 443:



💡 Using TLS filter:

Capturing from Wi-Fi 2

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Options: Narrow & Wide Case sensitive Backwards Multiple occurrences

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	18.204.36.77	192.168.29.83	TLsv1.2	118	Application Data
15	2.649840	2600:140f:7208:3c::	2405:201:3027:5047::	TLsv1.2	98	Application Data
19	2.691219	2600:140f:7208:3c::	2405:201:3027:5047::	TLsv1.2	98	Application Data
33	3.485945	2405:201:3027:5047::	2620:1ec:21::14	TLsv1.2	1331	Application Data
34	3.486212	2405:201:3027:5047::	2620:1ec:21::14	TLsv1.2	120	Application Data
35	3.486304	2405:201:3027:5047::	2620:1ec:21::14	TLsv1.2	443	Application Data
43	3.561569	2620:1ec:21::14	2405:201:3027:5047::	TLsv1.2	120	Application Data
47	3.770843	2620:1ec:21::14	2405:201:3027:5047::	TLsv1.2	255	Application Data
63	6.816287	2620:1ec:21::14	2405:201:3027:5047::	TLsv1.2	120	Application Data
64	6.821121	2405:201:3027:5047::	2620:1ec:21::14	TLsv1.2	116	Application Data
65	6.821473	2405:201:3027:5047::	2620:1ec:21::14	TLsv1.2	116	Application Data
97	12.745600	192.168.29.83	18.204.36.77	TLsv1.2	227	Application Data
111	13.029991	192.168.29.83	52.20.228.253	TLsv1.2	293	Application Data
112	13.063338	18.204.36.77	192.168.29.83	TLsv1.2	113	Application Data
115	13.301649	52.20.228.253	192.168.29.83	TLsv1.2	108	Application Data
116	13.302514	192.168.29.83	52.20.228.253	TLsv1.2	1091	Application Data
117	13.569219	52.20.228.253	192.168.29.83	TLsv1.2	225	Application Data
121	14.001990	192.168.29.83	44.214.32.211	TLsv1.2	366	Application Data
129	14.319956	44.214.32.211	192.168.29.83	TLsv1.2	108	Application Data
130	14.320358	192.168.29.83	44.214.32.211	TLsv1.2	1064	Application Data
131	14.525251	2404:6800:4002:823::	2405:201:3027:5047::	TLsv1.2	147	Application Data
135	14.634761	44.214.32.211	192.168.29.83	TLsv1.2	512	Application Data

> Frame 1: 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on interface \Device\NPF...
> Ethernet II, Src: ServercomPri_91:0f:ee (8c:a3:99:91:0f:ee), Dst: 6e:ec:b2:0c:54:09 (6e:ec:b2:0c:54:09)
> Internet Protocol Version 4, Src: 18.204.36.77, Dst: 192.168.29.83
> Transmission Control Protocol, Src Port: 443, Dst Port: 52205, Seq: 1, Ack: 1, Len: 64
> Transport Layer Security

Transport Layer Security: Protocol

Packets: 8647 - Displayed: 2209 (25.5%)

72°F Mostly cloudy

Search

ENG IN

10:22 PM 2/9/2025

Identify the IP address of a website you visit

Capturing from wlp0s20f3

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ip.src==192.168.42.127

No.	Time	Source	Destination	Protocol	Length	Info
8	741.1	192.168.42.127	140.82.112.26	TCP	54	60402 -> 443 [RST] Seq=1907 Win=0 Len=0
8	741.1	192.168.42.127	140.82.112.26	TCP	54	60402 -> 443 [RST] Seq=1907 Win=0 Len=0
8	689.0	192.168.42.127	185.125.190.18	TCP	74	46202 -> 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=445670229 TSecr=1403870...
8	689.7	192.168.42.127	185.125.190.18	TCP	66	46202 -> 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=445670832 TSecr=1403870...
8	689.7	192.168.42.127	185.125.190.18	HTTP	154	GET / HTTP/1.1
8	690.4	192.168.42.127	185.125.190.18	TCP	66	46202 -> 80 [ACK] Seq=89 Ack=190 Win=64128 Len=0 TSval=445671548 TSecr=1403...
8	690.4	192.168.42.127	185.125.190.18	TCP	66	46202 -> 80 [FIN, ACK] Seq=89 Ack=191 Win=64128 Len=0 TSval=445671548 TSecr=1403...
9	988.6	192.168.42.127	185.125.190.49	TCP	74	42992 -> 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=777565955 TSecr=2550368...
9	988.8	192.168.42.127	185.125.190.49	TCP	66	42992 -> 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=777565327 TSecr=2550368...
9	988.8	192.168.42.127	185.125.190.49	HTTP	154	GET / HTTP/1.1
9	989.3	192.168.42.127	185.125.190.49	TCP	66	42992 -> 80 [ACK] Seq=89 Ack=190 Win=64128 Len=0 TSval=777565782 TSecr=2550...
9	989.3	192.168.42.127	185.125.190.49	TCP	66	42992 -> 80 [FIN, ACK] Seq=89 Ack=191 Win=64128 Len=0 TSval=777565782 TSecr=2550...
1	129.0	192.168.42.127	192.168.42.186	DNS	86	Standard query 0x337d AAAA ssl.gstatic.com OPT
1	129.0	192.168.42.127	192.168.42.186	DNS	86	Standard query 0x9c2a A ssl.gstatic.com OPT
1	129.1	192.168.42.127	192.168.42.186	DNS	75	Standard query 0x139a AAAA ssl.gstatic.com
1	129.1	192.168.42.127	192.168.42.186	DNS	75	Standard query 0xaaba A ssl.gstatic.com
1	129.1	192.168.42.127	192.168.42.186	DNS	75	Standard query 0x2033 HTTPS ssl.gstatic.com
1	145.0	192.168.42.127	192.168.42.186	DNS	79	Standard query 0x337d HTTPS accounts.google.com
1	145.0	192.168.42.127	192.168.42.186	DNS	79	Standard query 0x7d5f A accounts.google.com
1	145.0	192.168.42.127	192.168.42.186	DNS	79	Standard query 0xa33c AAAA accounts.google.com
1	148.8	192.168.42.127	192.168.42.186	DNS	75	Standard query 0xadf2 A www.youtube.com
1	148.8	192.168.42.127	192.168.42.186	DNS	75	Standard query 0x4aed AAAA www.youtube.com
1	148.8	192.168.42.127	192.168.42.186	DNS	83	Standard query 0x6526 A youtube-ui.l.google.com
1	148.8	192.168.42.127	192.168.42.186	DNS	83	Standard query 0x5471 AAAA youtube-ui.l.google.com

> Frame 1360: 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface wlp0s20f3, id 0
> Ethernet II, Src: Intel_85:5c:a5 (3c:21:9c:85:5c:a5), Dst: 46:6d:94:af:05:ed (46:6d:94:af:05:ed)
> Internet Protocol Version 4, Src: 192.168.42.127, Dst: 192.168.42.186
> User Datagram Protocol, Src Port: 57851, Dst Port: 53
> Domain Name System (query)

wireshark wlp0s20f337JW02.pcapng

Packets: 9735 - Displayed: 228 (2.3%)

Profile: Default

Ping google.com:

```
Command Prompt
Request timed out.
Reply from 2404:6800:4002:817::200e: time=148ms
Reply from 2404:6800:4002:817::200e: time=56ms

Ping statistics for 2404:6800:4002:817::200e:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 32ms, Maximum = 148ms, Average = 78ms

C:\Users\HP>tracert google.com

Tracing route to google.com [2404:6800:4002:821::200e]
over a maximum of 30 hops:

  1    5 ms    4 ms    6 ms    2405:201:3027:5047:8ea3:99ff:fe91:fee
  2    *        *        *        Request timed out.
  3   10 ms    9 ms    8 ms    2405:200:801:1200::1004
  4    7 ms    8 ms   11 ms    2405:200:801:1200::105e
  5   52 ms   12 ms   14 ms    2405:200:870:3168:61::5
  6    *        *        *        Request timed out.
  7   15 ms   10 ms   12 ms    2405:200:801:1200::4bc
  8    *        *        *        Request timed out.
  9   42 ms   30 ms   30 ms    2405:200:801:200::31d
 10    *        *        *        Request timed out.
 11   25 ms   22 ms   27 ms    2404:6800:811c::1
 12   66 ms   47 ms   29 ms    2001:4860:0:1::3128
 13   34 ms    *        *        2001:4860:0:1::8760
 14   34 ms   34 ms   34 ms    2001:4860:9:4001:67bc
 15   84 ms   38 ms   34 ms    2001:4860:c:4004:172c
 16   36 ms   38 ms   38 ms    2001:4860:9:4001:ddce
 17   33 ms   35 ms   33 ms    2001:4860:0:1::78c1
 18   45 ms   43 ms   38 ms    2001:4860:0:1::53a3
 19   36 ms   32 ms   33 ms    dell2s04-in-x0e.1e100.net [2404:6800:4002:821::200e]

Trace complete.
```

Traceoute google.com:

```
C:\Users\HP>ping google.com

Pinging google.com [2404:6800:4002:817::200e] with 32 bytes of data:
Reply from 2404:6800:4002:817::200e: time=32ms
Request timed out.
Reply from 2404:6800:4002:817::200e: time=148ms
Reply from 2404:6800:4002:817::200e: time=56ms

Ping statistics for 2404:6800:4002:817::200e:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 32ms, Maximum = 148ms, Average = 78ms

C:\Users\HP>tracert google.com

Tracing route to google.com [2404:6800:4002:821::200e]
over a maximum of 30 hops:

  1    5 ms    4 ms    6 ms    2405:201:3027:5047:8ea3:99ff:fe91:fee
  2    *        *        *        Request timed out.
  3   10 ms    9 ms    8 ms    2405:200:801:1200::1004
  4    7 ms    8 ms   11 ms    2405:200:801:1200::105e
  5   52 ms   12 ms   14 ms    2405:200:870:3168:61::5
  6    *        *        *        Request timed out.
  7   15 ms   10 ms   12 ms    2405:200:801:1200::4bc
  8    *        *        *        Request timed out.
  9   42 ms   30 ms   30 ms    2405:200:801:200::31d
 10    *        *        *        Request timed out.
 11   25 ms   22 ms   27 ms    2404:6800:811c::1
 12   66 ms   47 ms   29 ms    2001:4860:0:1::3128
```

xlklopi